

Math 142, F20
Final Exam

Name: *Answer*

Instructions:

- There are totally 8 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	15	
2	14	
3	12	
4	15	
5	14	
6	14	
7	10	
8	12	
Total	106	

Good Luck !

Problem 1. (15 points) Evaluate the following integrals.

(a) $\int_0^1 \frac{e^{\arctan x}}{1+x^2} dx$ (Hint: Use appropriate substitution)

$$\begin{aligned} \text{Let } u &= \arctan x, \quad du = \frac{1}{1+x^2} dx \\ x=0 &\Rightarrow u = \arctan 0 = 0 \\ x=1 &\Rightarrow u = \arctan 1 = \pi/4 \\ \Rightarrow \int_0^1 \frac{e^{\arctan x}}{1+x^2} dx &= \int_0^{\pi/4} e^u du = e^u \Big|_{u=0}^{u=\pi/4} \\ &= e^{\pi/4} - e^0 = e^{\pi/4} - 1 \end{aligned}$$

(b) $\int x e^{-2x} dx$ (Hint: Use integration by parts)

$$\begin{aligned} u &= x, \quad dv = e^{-2x} dx \\ du &= dx, \quad v = -\frac{e^{-2x}}{2} \\ \Rightarrow \int x e^{-2x} dx &= x \left(-\frac{e^{-2x}}{2} \right) - \int -\frac{e^{-2x}}{2} dx \\ &= -\frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} + C \\ &= -\frac{1}{4}(2x+1)e^{-2x} + C \end{aligned}$$

(c) $\int \sin^3 \theta \cos^2 \theta d\theta$ (Hint: Use appropriate substitution)

$$\begin{aligned} &= \int \sin \theta \sin^2 \theta \cos^2 \theta d\theta \\ &= \int \sin \theta (1 - \cos^2 \theta) \cos^2 \theta d\theta \quad \leftarrow \text{let } u = \cos \theta \\ &\quad \quad \quad du = -\sin \theta d\theta \\ &= \int (1 - u^2) u^2 (-du) \\ &= \int -u^2 + u^4 du = -\frac{u^3}{3} + \frac{u^5}{5} + C \\ &= -\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5} + C \end{aligned}$$

Problem 2. (14 points) Evaluate the following integrals.

(a) $\int \frac{1}{t^2 \sqrt{9-t^2}} dt$

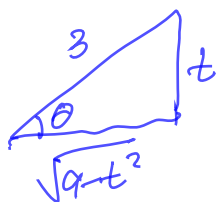
(Hint: Use appropriate trigonometric substitution)

$$\text{let } t = 3\sin\theta \Rightarrow \sqrt{9-t^2} = 3\cos\theta$$

$$dt = 3\cos\theta d\theta$$

$$\int \frac{1}{t^2 \sqrt{9-t^2}} dt = \int \frac{1}{(3\sin\theta)^2 3\cos\theta} 3\cos\theta d\theta$$

$$= \int \frac{1}{9\sin^2\theta} d\theta = \int \frac{1}{9} \csc^2\theta d\theta$$



$$= -\frac{1}{9} \cot\theta + C$$

$$= -\frac{1}{9} \cdot \frac{\sqrt{9-t^2}}{t} + C$$

(b) $\int \frac{x-8}{x^2+x-6} dx$

(Hint: Use method of partial fractions)

$$\frac{x-8}{x^2+x-6} = \frac{x-8}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2} = \frac{A(x-2)+B(x+3)}{(x+3)(x-2)}$$

$$= \frac{(A+B)x + (-2A+3B)}{(x+3)(x-2)} \Rightarrow \begin{cases} A+B=1 \\ -2A+3B=-8 \end{cases} \Rightarrow \begin{matrix} A=1/5 \\ B=-6/5 \end{matrix}$$

$$\Rightarrow \int \frac{x-8}{x^2+x-6} dx = \int \frac{1}{5} \cdot \frac{1}{x+3} - \frac{6}{5} \frac{1}{x-2} dx$$

$$= \frac{1}{5} \ln|x+3| - \frac{6}{5} \ln|x-2| + C$$

Problem 3. (12 points) Evaluate each of the following improper integral or show that it is divergent.

(a) $\int_0^{+\infty} \sin(2x) dx$

$$= \lim_{b \rightarrow +\infty} \int_0^b \sin(2x) dx$$

$$= \lim_{b \rightarrow +\infty} -\frac{1}{2} \cos 2x \Big|_{x=0}^{x=b}$$

$$= \lim_{b \rightarrow +\infty} -\frac{1}{2} \cos 2b + \frac{1}{2} \rightarrow \text{DNE}$$

$$\Rightarrow \text{diverges.}$$

(b) $\int_0^2 y^2 \ln y dy$

$$\text{let } u = \ln y, dv = y^2 dy$$

$$du = \frac{1}{y} dy, v = \frac{y^3}{3}$$

$$\Rightarrow \int y^2 \ln y dy = \frac{1}{3} y^3 \ln y - \int \frac{y^3}{3} \cdot \frac{1}{y} dy$$

$$= \frac{1}{3} y^3 \ln y - \int \frac{y^2}{3} dy = \frac{y^3 \ln y}{3} - \frac{y^3}{9} + C$$

$$\Rightarrow \int_0^2 y^2 \ln y dy = \lim_{a \rightarrow 0} \int_a^2 y^2 \ln y dy = \lim_{a \rightarrow 0} \left(\frac{y^3 \ln y}{3} - \frac{y^3}{9} \right) \Big|_{y=a}^{y=2}$$

$$= \lim_{a \rightarrow 0} \left(\frac{8 \ln 2}{3} - \frac{8}{9} \right) - \left(\frac{a^3 \ln a}{3} - \frac{a^3}{9} \right)$$

$$= \frac{8 \ln 2}{3} - \frac{8}{9}$$

Problem 4. (15 points) Determine whether each of the following series is convergent or divergent.

(a) $\sum_{n=1}^{+\infty} \frac{1+3^n}{4^n}$ (If convergent, also find its sum for this one)

$$\begin{aligned}
 &= \sum_{n=1}^{+\infty} \frac{1}{4^n} + \sum_{n=1}^{+\infty} \left(\frac{3}{4}\right)^n \rightarrow \text{Converges by geometric series Property.} \\
 &= \frac{\frac{1}{4}}{1-\frac{1}{4}} + \frac{\frac{3}{4}}{1-\frac{3}{4}} \\
 &= \frac{\frac{1}{4}}{\frac{3}{4}} + \frac{\frac{3}{4}}{\frac{1}{4}} = \frac{1}{3} + 3 = \frac{10}{3}
 \end{aligned}$$

(b) $\sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}} \Rightarrow f(x) = \frac{1}{x\sqrt{\ln x}}$ (Hint: Use Integral Test)

$$\begin{aligned}
 \int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx &= \left[2\sqrt{\ln x} \right]_{x=2}^{x=+\infty} \\
 &= \lim_{b \rightarrow +\infty} [2\sqrt{\ln b} - 2\sqrt{\ln 2}] = +\infty
 \end{aligned}$$

$$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}} \text{ diverges by the Integral Test}$$

(c) $\sum_{n=1}^{+\infty} \frac{\sqrt{4n+1}}{2n^2+n+2}$ (Hint: Use Limit Comparison Test)

$$a_n = \frac{\sqrt{4n+1}}{2n^2+n+2} \quad b_n = \frac{\sqrt{4n}}{2n^2} = \frac{1}{n^{3/2}}$$

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = 1$$

$$\begin{aligned}
 \sum_{n=1}^{+\infty} b_n &= \sum_{n=1}^{+\infty} \frac{1}{n^{3/2}} \text{ Converges} \\
 \left. \begin{aligned} &\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = 1 \\ &\sum_{n=1}^{+\infty} b_n \text{ Converges} \end{aligned} \right\} &\Rightarrow \sum_{n=1}^{+\infty} \frac{\sqrt{4n+1}}{2n^2+n+2} \text{ Converges by the Limit Comparison Test.}
 \end{aligned}$$

Problem 5. (14 points) Find the radius of convergence and interval of convergence for each of the following series.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{4^n n^2}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1}}{4^{n+1} (n+1)^2} \cdot \frac{4^n n^2}{(-1)^n x^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n^2}{(n+1)^2} \cdot \frac{x}{4} \right| = \frac{|x|}{4} < 1 \Rightarrow \underline{R=4}$$

check the end points $x=4, -4$

i) $x=4, \sum_{n=1}^{\infty} \frac{(-1)^n 4^n}{4^n n^2} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ Converges

ii) $x=-4, \sum_{n=1}^{\infty} \frac{(-1)^n (-4)^n}{4^n n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}$ Converges

$\Rightarrow$ Interval of Convergence: $[-4, 4]$

(b) $\sum_{n=1}^{+\infty} \frac{3^n (x+2)^n}{\sqrt{n}}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^{n+1} (x+2)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{3^n (x+2)^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{n}}{\sqrt{n+1}} \cdot 3(x+2) \right| = 3|x+2| < 1$$

$$\Rightarrow |x+2| < \frac{1}{3} \Rightarrow \underline{R=\frac{1}{3}}$$

check the end points: $-2 + \frac{1}{3} = -\frac{5}{3}$ & $-2 - \frac{1}{3} = -\frac{7}{3}$

1) $x = -\frac{5}{3}, \sum_{n=1}^{\infty} \frac{3^n \left(-\frac{1}{3}\right)^n}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges

2) $x = -\frac{7}{3}, \sum_{n=1}^{\infty} \frac{3^n \left(-\frac{1}{3}\right)^n}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converges

$\Rightarrow$ Interval of Convergence: $\left[-\frac{7}{3}, -\frac{5}{3}\right)$

Problem 6. (14 points)

(a) Determine whether the series is conditionally convergent, absolutely convergent or divergent:

$$\sum_{n=1}^{+\infty} \frac{(-1)^n \sin(1/n)}{n}$$

let $a_n = \frac{\sin(\frac{1}{n})}{n}$, $b_n = \frac{1}{n^2}$ ($\sin(\frac{1}{n}) > 0$ for any n)

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \lim_{n \rightarrow +\infty} \frac{\sin(\frac{1}{n})}{n} \cdot n^2 = \lim_{n \rightarrow +\infty} n \cdot \sin(\frac{1}{n}) = 1.$$

$$\sum_{n=1}^{+\infty} \frac{1}{n^2} \text{ converges} \Rightarrow \sum_{n=0}^{+\infty} \frac{\sin(\frac{1}{n})}{n} \text{ converges}$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^n \sin(\frac{1}{n})}{n} \text{ absolutely converges}$$

(b) Find the Maclaurin series (power series centered at 0) and its radius of convergence for the function $f(x) = \frac{x}{2+x^2}$. (Hint: Use $\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n$ for $|x| < 1$)

$$f(x) = x \cdot \frac{1}{2+x^2} = \frac{x}{2} \cdot \frac{1}{1+(\frac{x^2}{2})} = \frac{x}{2} \cdot \frac{1}{1-(-\frac{x^2}{2})}$$

$$= \frac{x}{2} \sum_{n=0}^{+\infty} \left(-\frac{x^2}{2}\right)^n = \frac{x}{2} \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{2^n}$$

$$= \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{2^{n+1}}, \quad \left| \frac{x^2}{2} \right| < 1 \Rightarrow |x^2| < 2 \Rightarrow |x| < \sqrt{2}$$

$$\Rightarrow R = \sqrt{2}$$

Problem 7. (10 points)

(a) Find an equation for the line tangent to the parametrized curve $x = 1/t$, $y = -2 + \ln t$ at the point defined by $t = 1$.

$$x = \frac{1}{t} \Rightarrow \frac{dx}{dt} = -\frac{1}{t^2} \Rightarrow m = \left. \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \right|_{t=1} = \left. \frac{\frac{1}{t}}{-\frac{1}{t^2}} \right|_{t=1} = -1$$
$$y = -2 + \ln t \Rightarrow \frac{dy}{dt} = \frac{1}{t}$$

At $t=1$, we have $x=1$, $y=-2$

$\Rightarrow$ The tangent line is

$$y + 2 = (-1)(x - 1)$$

$$y = -x - 1$$

(b) Find the polar coordinates (r, θ) of the point with Cartesian coordinates $(-1, -\sqrt{3})$, where $r > 0$ and $0 \leq \theta < 2\pi$.

$$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{y}{x} = \frac{-\sqrt{3}}{-1} = \sqrt{3}$$

$(-1, -\sqrt{3})$ is in the third quadrant $\Rightarrow \theta = \frac{4}{3}\pi$.

Thus the polar coordinates are

$$(2, \frac{4\pi}{3})$$

Problem 8. (12 points) Given a polar equation $r = 2 \sin \theta$.

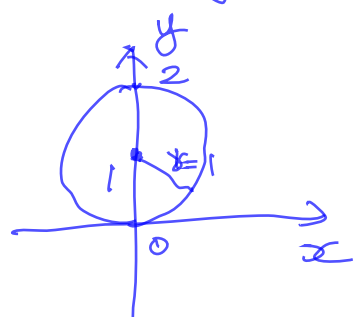
(a) Identify and sketch the curve by converting it to a Cartesian equation.

$$r = 2 \sin \theta \Rightarrow r^2 = 2 r \sin \theta.$$

$$\Leftrightarrow \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y = 0$$

$$x^2 + (y - 1)^2 = 1$$


(b) The area of the region that is bounded by this polar curve and lies in the sector $0 \leq \theta \leq \pi/3$ is given by the integral $A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta := \frac{1}{2} \int_0^{\pi/3} (2 \sin \theta)^2 d\theta$. Evaluate this integral to find the area.

$$A = \frac{1}{2} \int_0^{\pi/3} (2 \sin \theta)^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/3} 4 \sin^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/3} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \int_0^{\pi/3} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \left[\theta - \frac{1}{2} \sin 2\theta \right]_{\theta=0}^{\theta=\pi/3}$$

$$= \left[\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right] - 0$$

$$= \frac{\pi}{3} - \frac{\sqrt{3}}{4}$$